

KARTHIK BALAJI B M

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PROFILE SUMMARY

Passionate IT student with a strong background in designing secure systems and developing efficient algorithms. Proficient in Python, C, DBMS, and networking, I have a proven record of crafting innovative, data-driven projects that address real-world challenges. Committed to leveraging cutting-edge technologies to drive sustainable digital transformation and measurable impact.

WORK EXPERIENCE

Green Tamil Nadu Mission - Tamil Nadu Forest Department

Website Data Management Intern

Apr 2023 – May 2023

Chennai, Tamil Nadu

- Created a comprehensive user manual for the web portal, enhancing usability and reducing user queries by 30%.
- Streamlined data sorting and formatting processes for the Agro-department, improving data processing speed.
- Developed engaging infographics for presentations, boosting stakeholder engagement and communication clarity.

PROJECTS AND PUBLICATIONS

Senti-Meter: Emotion-Based Music Recommendation System | Python, OpenCV [GitHub Link](#) [Publication Proof](#)

Sep 2024

- Developed a real-time emotion-based music recommendation system, improving user engagement by 30%. Achieved 92% accuracy in emotion detection.
- Built a responsive React.js frontend for real-time interaction, with a Flask backend for seamless data processing. Achieved high accuracy in emotion detection, supporting SDG goals for health, innovation, and sustainable communities.
- Published work as 'Real-Time Emotion-Based Music Recommendation System Using YOLOv5 and Spotify API', contributing to personalized music and sustainable community development.

Know-the-Weather: Real-Time Weather Forecasting Web Application | React.js, OpenWeatherMap API, CSS [GitHub Link](#)

Sep 2023

- Developed a responsive web application to fetch and display real-time weather information using the OpenWeatherMap API.
- Built the frontend using React.js with reusable components and implemented efficient API handling.
- Integrated environment variables for secure handling of API keys, ensuring scalability and security.
- Enhanced user experience with responsive design, providing seamless functionality across devices.

Understanding AI Model Collapse: Causes, Impacts, and Solutions [Publication Proof](#)

ICSTSDG 2024

- Made a study on AI model collapse, including data challenges and biases. Proposed mitigation strategies to improvement in model stability.

SKILLS

- Programming Languages & Libraries:** Python, C, SQL, Java, NumPy, Pandas, Matplotlib
- Tools & Software:** GitHub, VS Code, Figma, Colab, PyCharm
- Technical Expertise:** Data Structures and Algorithms, Object-Oriented Programming (OOPs)
- Soft Skills:** Solution-Oriented, Critical Thinking, Adaptability, Emotional Intelligence

EDUCATION

Sri Sairam Engineering College

Bachelor of Technology in Information Technology (IT) **CGPA: 7.69**

July 2021 – Present

Chennai, Tamil Nadu

ACHIEVEMENTS & CO-CURRICULAR

Awards: Received the **Elite Performer Award** for excellence in technical events, intercollegiate contests, and interviews [LinkedIn Post](#).

LeetCode: Solved over **400+** problems, strengthening DSA skills. [LeetCode Profile](#).

Hackathons: Won the **Infoculture Fiesta** organized by NDLI Club with over 100+ participants [Certificate](#).

Coding Contests: Secured top positions among 500+ participants in national symposiums [Certificate](#).

Cybersecurity Contests: Competed in **Yukthi CTF 2024**, ranking among 150+ teams [Certificate](#).

Design: Finalist in the **Can-fig Web Design Contest** [Certificate](#).

Co-Curricular:

- [Google Intro to Large Language Models](#)
- [IBM Python for Data Science](#)
- [Cisco Introduction to Cybersecurity](#)